# Machine Learning

## Introduction




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https://www.aparat.com/mehran.safayani


https://github.com/safayani/machine_learning_course

https://git.iut.ac.ir/safayani/courses/-/tree/main/Machine_Learning_course


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# What is Machine Learning?

- **Learning from Data Instead of Writing Rules**
- **Traditional Programming:**
  - Data + Hand−written Rules ⟶ Output
  - We tell the computer exactly *how* to solve the problem step by step.
- **Machine Learning:**
  - Data + Correct Answers ⟶ Learned Rules (Model)
  - The computer discovers the rules on its own by looking at examples.
- **Why it matters:**
  - Gives computers a new ability: solving problems that are too complex to code by hand.







# Why Hand-Written Code Fails (Example 1)

- **Why Hand-Written Rules Fail: Complex Vision**
- **Subtitle: Recognizing Objects in the Real World**
- **The Problem:**
  - Recognizing a 3D object from a new angle, under new lighting, or with background clutter.
- **Why can’t we just write if-else rules?**
  - **We don’t know our own rules:** Our brain does it instantly, but we cannot explain the exact steps in code.
  - **Too many edge cases:** Shadows, rotations, and partial views create endless variations.
  - **Pixels are deceptive:** Raw pixel numbers change completely with a slight shift in light, even if the object stays the same.
- **Result:** Any hand-written rule system becomes impossibly long, messy, and fragile.

# Why Hand-Written Code Fails (Example 2)

- **Why Hand-Written Rules Fail: Moving Targets**
- **Subtitle: Detecting Credit Card Fraud**
- **The Problem:**
  - Deciding whether a credit card transaction is real or fraudulent.
- **Why rules don’t work here:**
  - **No single rule works:** There is no simple check (e.g., “amount > $5,000”) that is reliable on its own.
  - **Combining weak clues:** Legitimate detection requires checking dozens of small clues at the same time (location, time, habit, device).
  - **Fraud is a moving target:** As soon as you write a rule, fraudsters figure it out and change their behavior.
- **Result:** Hand-written rules become outdated almost immediately.

# The Machine Learning Approach

- **Collect Examples →Train an Algorithm →Generalize**
- **Step 1: Collect Examples:**
  - Gather inputs (images, transactions) paired with correct labels (digit label, fraud / not fraud).
- **Step 2: Train the Algorithm:**
  - A learning algorithm searches through parameters (weights) to find the best fit for the data.
  - The resulting “program” is a mathematical model, often composed of thousands or millions of numbers.
- **Step 3: Generalize to New Cases:**
  - A good model does not just memorize training examples—it works on **unseen data**.
- **Step 4: Adapt Easily:**
  - When patterns change, we don’t rewrite code from scratch; we simply **retrain the model on new data**.
- **Economic Fact:** Running training on hardware is far cheaper and faster than paying humans to maintain thousands of fragile rules.

# Standard Example: MNIST

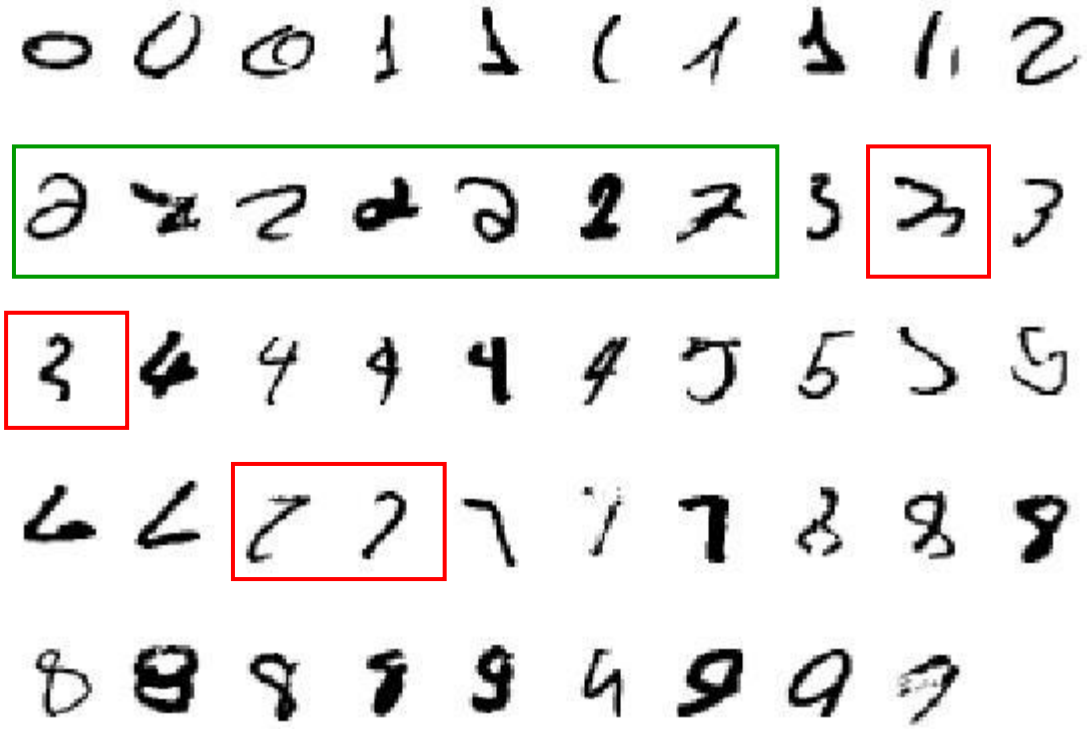
- **What is MNIST?**
  - A classic dataset of 70,000 handwritten digit images ($0$ to $9$).
  - Image size: $28 \times 28$pixels in grayscale ($784$ numbers per image).
- **Why is recognizing digits hard?**
  - People write numbers differently: slants, thickness, loops, and broken lines.
  - A simple template match does not work.
- **Our Plan:**
  - We will use MNIST throughout the course to compare different methods (Linear Classifiers, KNN, SVM, Decision Trees).







# It is very hard to say what makes a 2



# Classification: Cats vs. Dogs

- In classification, we want to assign each input to one of two classes.
- Here, each image is converted into a feature vector $x_i \in \mathbb{R}^d$.
- The goal is to learn a rule that separates the classes using the training data.
- In this example, red points represent one class and blue points represent the other class.
- A **linear classifier** separates the two classes using a straight line.
- New examples are classified according to which side of the line they fall on.
- This is the basic idea behind many machine learning classifiers.



# Classification

$\boldsymbol{x}_i \in \mathbb{R}^d$

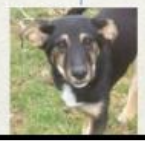


Training data





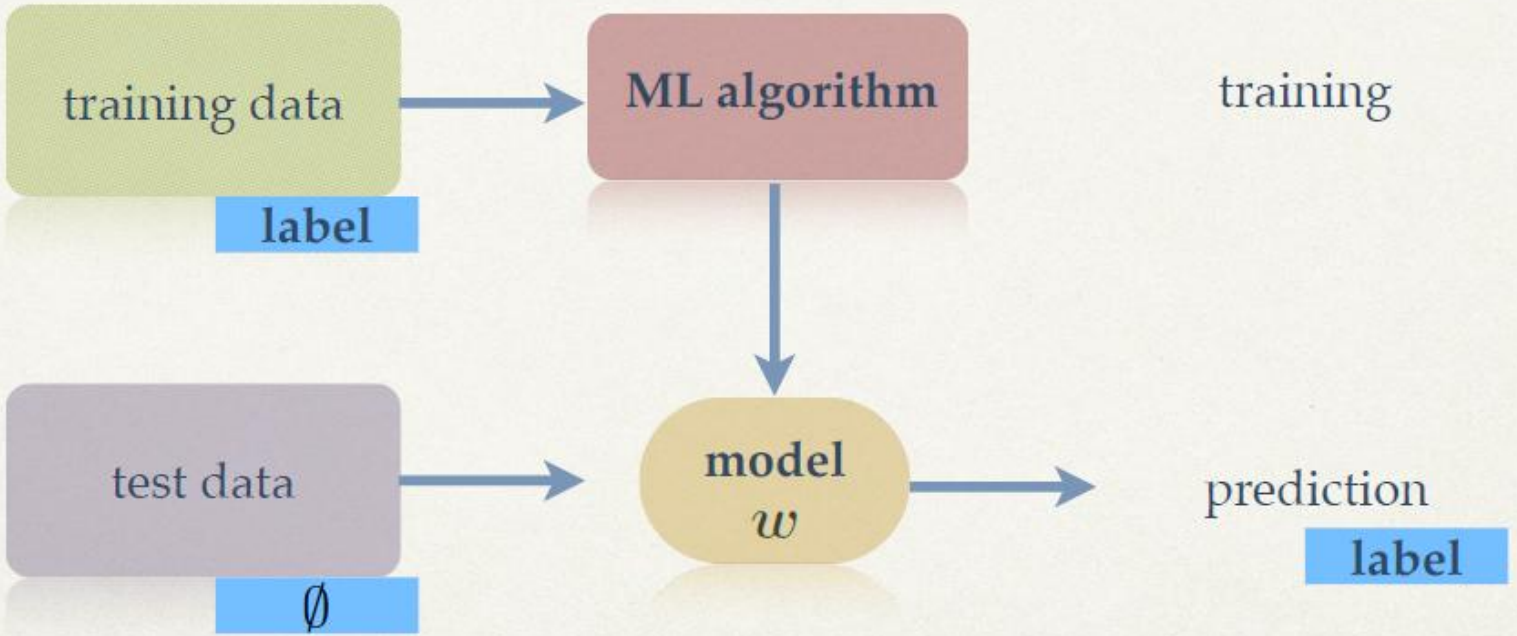








# Machine Learning Fundamentals

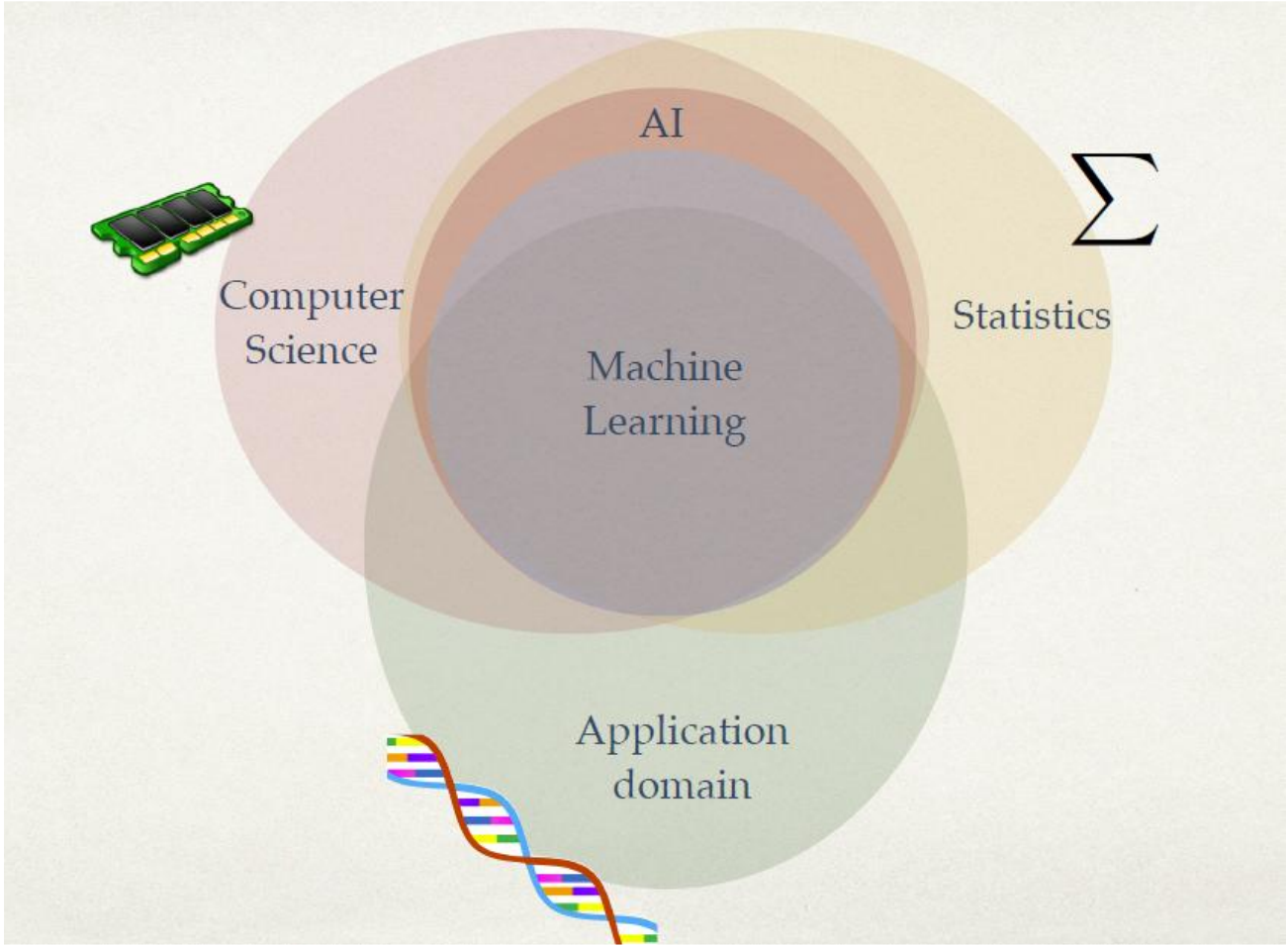


# Machine Learning definition

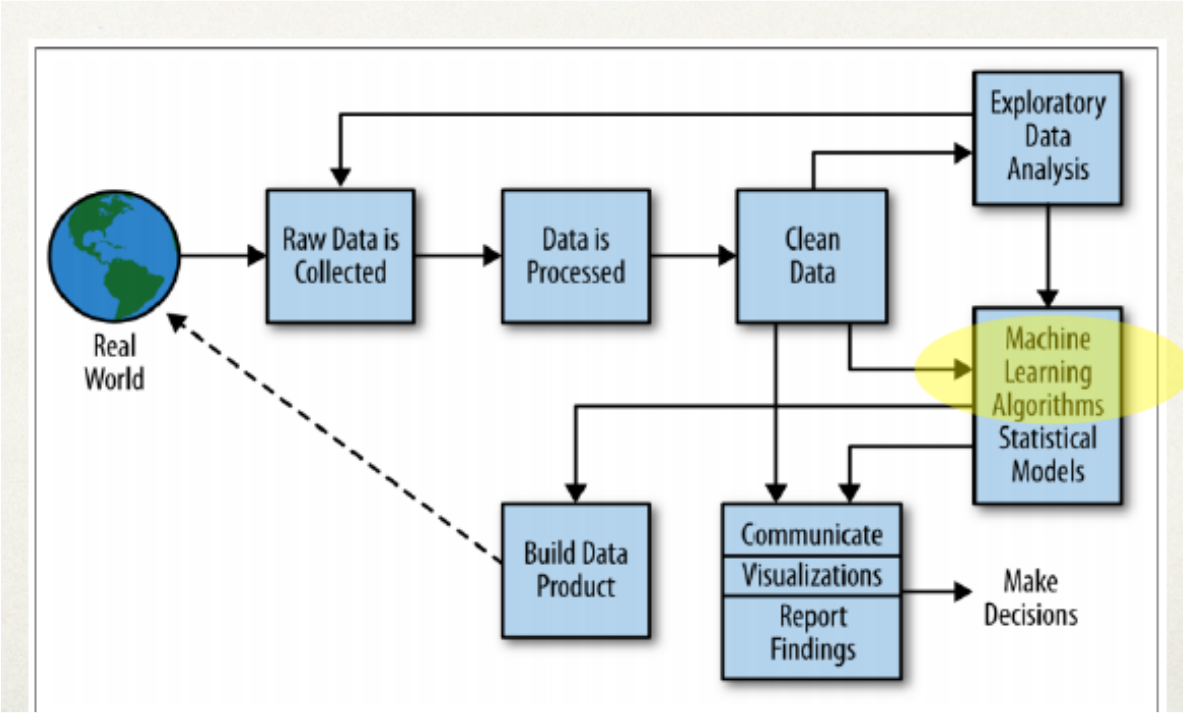
- "Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed."
  — *Arthur Samuel, 1959*






AI
Computer Science
Statistics
Machine Learning
Application domain

# ML is only a small part!



More Examples:

- Database mining

  Large datasets from growth of automation/web.

  E.g., Web click data, medical records, biology, engineering

- Applications can’t program by hand.

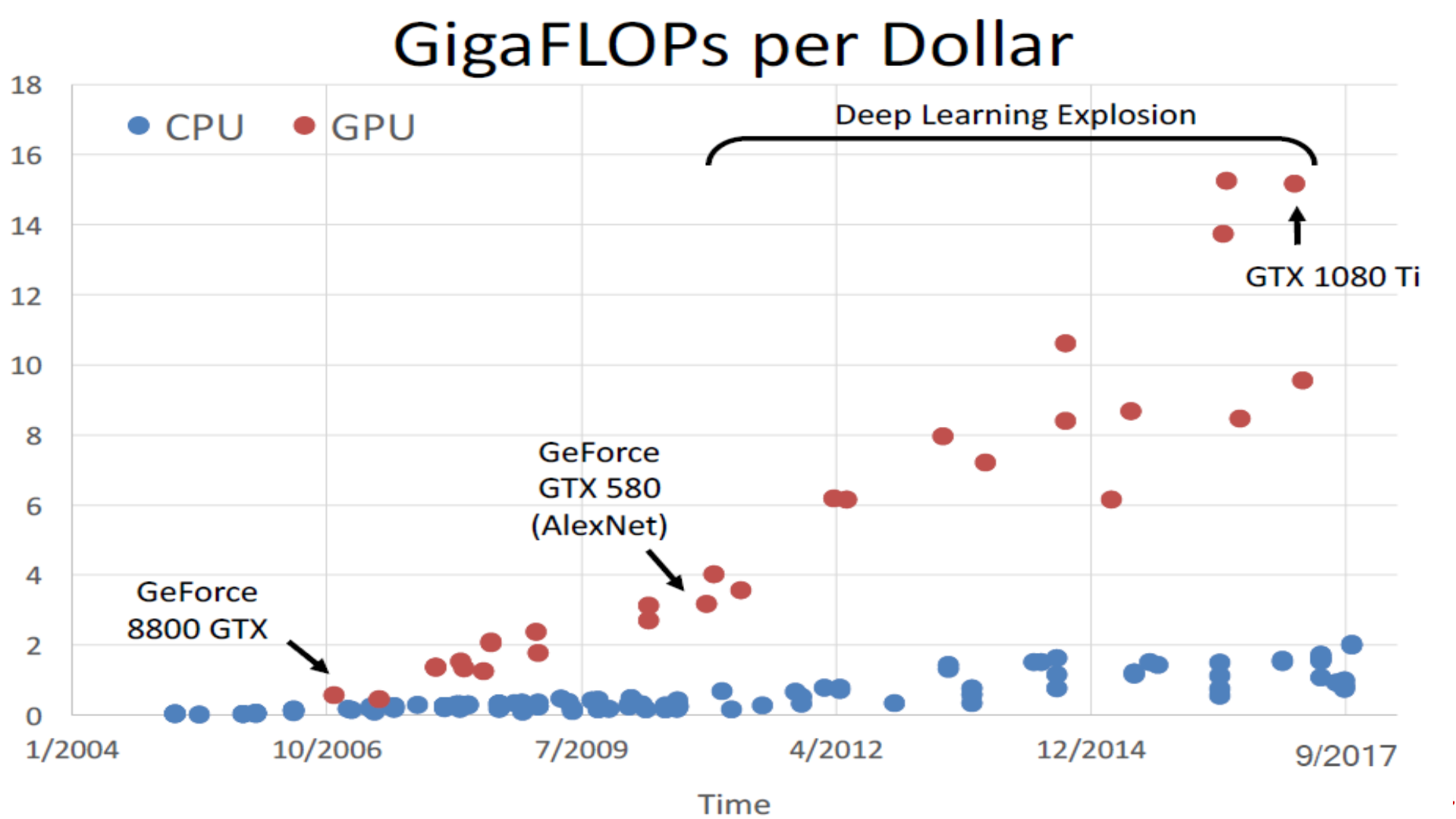
  E.g., Autonomous helicopter, handwriting recognition, most of Natural Language Processing (NLP), Computer Vision.

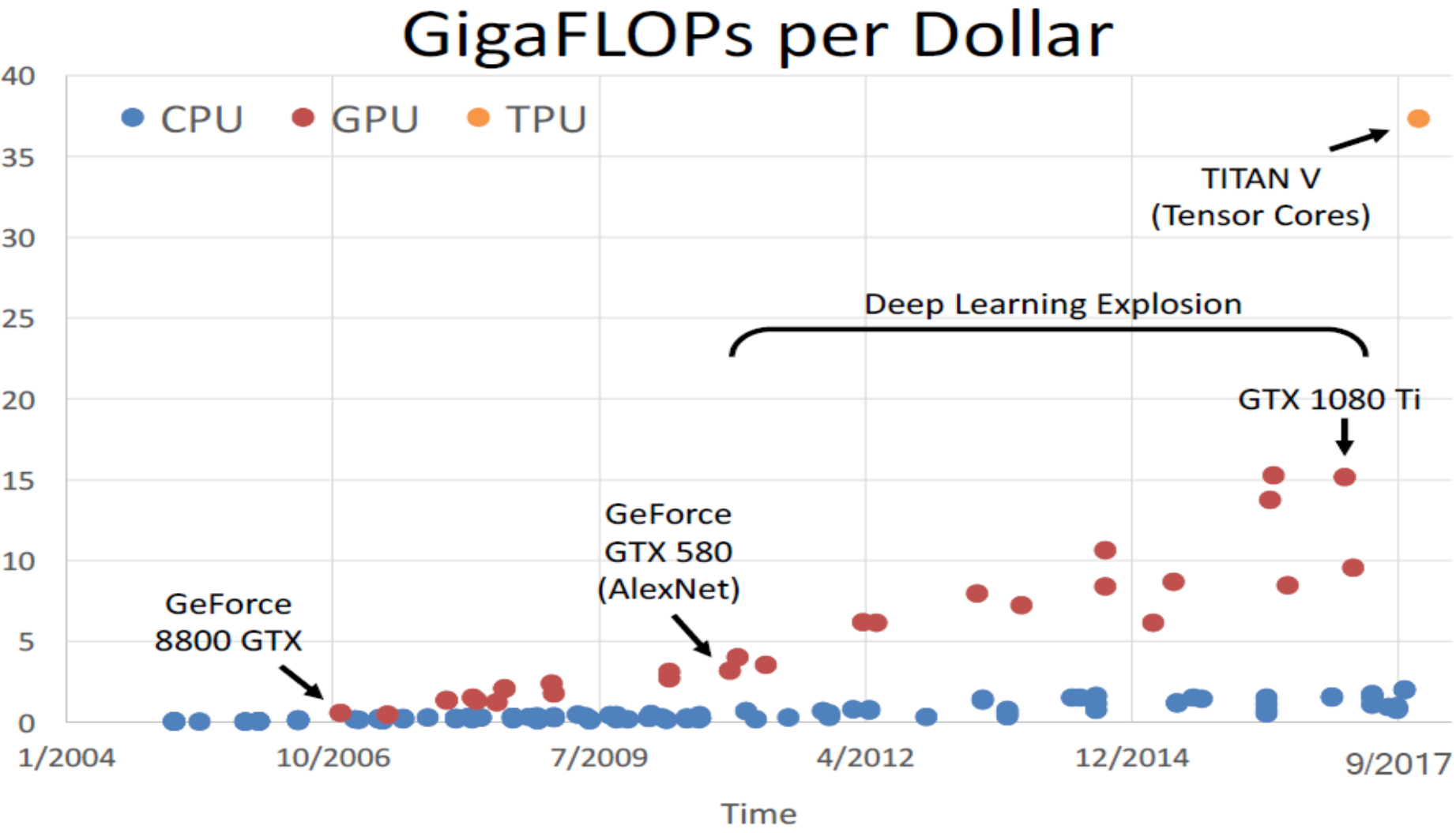
- Self-customizing programs

  E.g., Amazon, Netflix product recommendations






GigaFLOPs per Dollar
CPU
GPU
Deep Learning Explosion
GTX 1080 Ti
GeForce
GTX 580
(AlexNet)
GeForce
8800 GTX
18
16
14
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6
4
2
0
1/2004
10/2006
7/2009
4/2012
12/2014
9/2017
Time


GigaFLOPs per Dollar
CPU
GPU
TPU
40
35
30
25
20
15
10
5
0
TITAN V
(Tensor Cores)
Deep Learning Explosion
GTX 1080 Ti
GeForce
GTX 580
(AlexNet)
GeForce
8800 GTX
1/2004
10/2006
7/2009
4/2012
12/2014
9/2017
Time

# What to expect?

- overview over ML
- basic understanding of most important ML methods and fundamental concepts
- experience how ML is done on a practical problem





# What not to expect?

- instantly becoming an **expert** data scientist / ML researcher / statistician / large-scale ML specialist / ML software expert / domain expert
- You will not learn ALL advanced methods.
- This course is not specially about big data or large-scale methods.





# Course Syllabus: Machine Learning (Supervised & Data-Centric Foundations)

- **Chapter 1: Foundations, Optimization, and Linear Regression**
- **Introduction to Machine Learning:** Problem formulation, taxonomy, and end-to-end ML workflows.
- **Linear Regression & Ordinary Least Squares (OLS):**
  - Analytical solution via the Normal Equation.
  - Numerical optimization: Batch Gradient Descent, Mini-batch, and Stochastic Gradient Descent (SGD).
- **Probabilistic Perspective:** Maximum Likelihood Estimation (MLE) and Maximum A Posteriori (MAP).
- **Generalization & Model Selection:**
  - Overfitting vs. Underfitting.
  - Theoretical breakdown of the Bias-Variance Decomposition.
  - Cross-validation strategies: K-Fold and Stratified K-Fold.
- **Regularization:**
  - Ridge ($L_2$) and Lasso ($L_1$) regularization; geometric analysis of parameter shrinkage and sparsity.





# Course Syllabus: Machine Learning (Supervised & Data-Centric Foundations)

- **Chapter 2: Classification Foundations**
- **Logistic Regression:**
  - The Sigmoid link function, Cross-Entropy loss formulation, and gradient-based optimization.
- multiclass classification problems.
- **K-Nearest Neighbors (KNN):**
- Core classification algorithm, hyperparameter selection ($k$), Curse of Dimensionality





# Course Syllabus: Machine Learning (Supervised & Data-Centric Foundations)

- **Chapter 3: Margin-Based and Kernel Methods**
- **Support Vector Machines (SVM):**
  - Hard-margin SVM, constrained optimization, and Lagrangian Duality.
  - Soft-margin SVM and slack variables ($\xi_i$).
- **Kernel Methods & The Kernel Trick:**
  - Implicit non-linear feature mapping: $K(x, x') = \langle \phi, (x)\phi(x') \rangle$.
  - Common kernel functions: Polynomial and Radial Basis Function (RBF/Gaussian), with geometric interpretation.
  - Dual optimization
  - Support vector regression (SVR)





# Course Syllabus: Machine Learning (Supervised & Data-Centric Foundations)

- **Chapter 4: Tree-Based Models, Advanced Regression, and Ensembles**
- **Decision Trees (Classification & Regression):**
  - Node splitting criteria: Entropy, Information Gain, and Gini Impurity.
  - Regression Trees using Mean Squared Error (MSE) minimization.
  - Controlling model complexity: Cost-Complexity Pruning ($c_\alpha$).
  - Tree Regressors
- **Ensemble Learning:**
  - Theoretical rationale of ensemble methods for variance and bias reduction.
  - **Bagging**
  - **Boosting**





# Course Syllabus: Machine Learning (Supervised & Data-Centric Foundations)

- **Chapter 5: Data-Centric Challenges, Anomalies, and Limited Supervision**
- **Evaluation in Severely Imbalanced Settings:**
  - Precision-Recall curves vs. ROC-AUC.
- **Anomaly Detection:**
- **Semi-Supervised Learning (SSL)**
- **Active Learning (AL)**





# Prerequisites

- Statistics and probability
- Linear Algebra
- Programming





# Recommended Textbooks

- G. James, D. Witten, T. Hastie and R. Tibshirani: An introduction to statistical learning
- C. Bishop: Pattern Recognition and Machine Learning
- K. Murphy: Machine Learning: A Probabilistic Perspective





# Related Courses

- Machine Learning, EPFL, https://www.epfl.ch/labs/mlo/machine-learning-cs-433/

- Machine Learning, Standford university, Coursera https://www.coursera.org/learn/machine-learning





# Course Strategy

- Assignments and projects 20% to 30%
- Midterm and quizzes 30% to 40%
- Final and quizzes 30% to 40%








هوش مصنوعی برای همه

چیدمان صفحه

تنظیمات

۸۹۳ دنبال‌کننده

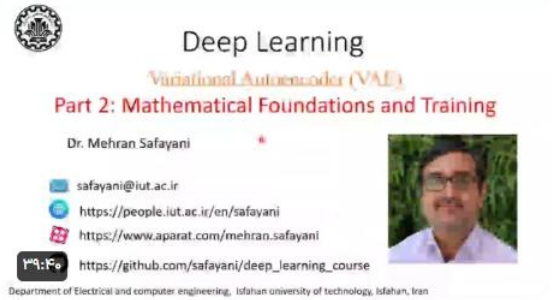
۸۱.۲ هزار بازدید کل

خانه | ویدیوها | ویدیوهای کوتاه | لیست‌های پخش | درباره کانال

یادگیر عمیق ج ۲۷ خودکدگذار تغییراتی (VAE) بخش ۲: مبانی ریاضی و آموزش ۱۴۰۵

چگونه مدل VAE را آموزش دهیم روابط آموزش VAE محاسبه ELBO Evidence Lower Bound تابع هدف حقه بازپارامترسازی Reparameterization trick مثال هایی از تولید داده

۱۲۷ بازدید ۴ ماه پیش




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